Research Square

Preprints are preliminary reports that have not undergone peer review.
They should not be considered conclusive, used to inform clinical practice,
or referenced by the media as validated information.

PTRR: A Metacognitive Framework for Measuring and Mitigating Automation Bias in AI-Assisted Vulnerability Research

Ziad Salah

daringosztafio@gmail.com

Ashraf A. Mohamed

Elsewedy University of Technology

Article

Keywords:

Posted Date: April 21st, 2026

DOI: https://doi.org/10.21203/rs.3.rs-9247251/v1

License: This work is licensed under a Creative Commons Attribution 4.0 International License.
Read Full License

Additional Declarations: No competing interests reported.

PTRR: A Metacognitive Framework for Measuring and Mitigating Automation Bias in AI-Assisted Vulnerability Research

Ziad Salah[1,*] **and Ashraf A. Mohamed**[2]

[1]Independent Security Researcher, Giza, Egypt
[2]Computer Engineering Department, Elsewedy University of Technology, Cairo, Egypt
[*]dariangosztafio@gmail.com

ABSTRACT

Artificial intelligence is increasingly integrated into professional cybersecurity workflows, yet the cognitive effects of AI assistance on researcher behavior remain poorly understood. This paper introduces the PTRR Framework, a structured metacognitive instrument designed to measure and mitigate Automation Bias in AI-assisted vulnerability research. PTRR comprises four components: Prompts (interaction quality scoring), Time (temporal behavioral analysis), Results (outcome-based severity scoring), and Rubric (process independence and verification criteria). Three derived indices operationalize the core constructs: the Automation Bias Index (ABI), which quantifies uncritical reliance on AI output; the Cognitive Struggle Index (CSI), which measures productive independent effort; and the Tool Integration Intensity Score (TIIS), which captures deliberate multi-tool synthesis. The framework is grounded in Dual Process Theory and Cognitive Load Theory, which together predict that expertise moderates the relationship between AI interaction quality and research output. A preliminary single-participant case study conducted over 60 structured hours provides initial construct validity evidence. The case study documented a shift from a low-to-medium severity finding profile to a critical-dominant profile following PTRR-based workflow adoption, including seven critical-severity findings validated by independent program triage and three multi-layer escalation chains. Total accepted findings increased from 2 to 11 over equivalent time periods. These results motivate the formal multi-participant study design proposed here as future work, which employs a within-subject Phase A versus Phase B design, Hierarchical Linear Modeling, and independent human scoring with pre-defined inter-rater reliability thresholds.

Introduction

The integration of large language models (LLMs) into professional security research workflows has accelerated substantially in recent years[1]. AI-assisted tools now support tasks spanning reconnaissance, vulnerability hypothesis generation, payload crafting, and report drafting—with recent work demonstrating that LLMs can be applied directly to automated penetration testing[4]. While this integration carries demonstrated efficiency benefits, it simultaneously introduces a well-documented class of systemic risks, including misplaced trust and uncritical acceptance of machine-generated output[6]. In the context of vulnerability research, these risks coalesce around *Automation Bias*—the tendency to preferentially accept machine-generated output without sufficient critical scrutiny[2,14]. A recent systematic review of the human-AI collaboration literature confirms that this bias is a pervasive and reproducible phenomenon across application domains[18].

The risk is not symmetric across practitioner populations. Experienced researchers who possess robust domain schemas and independent verification habits may benefit from AI as a cognitive amplifier. Novice practitioners, whose domain representations are still forming, face the hazard that AI assistance short-circuits the productive cognitive struggle necessary for deep skill acquisition[3]. This divergence constitutes what the present work terms the *Dual-Edge Effect*: the same level of AI assistance produces qualitatively different cognitive outcomes depending on the practitioner's expertise tier. Recent empirical work directly corroborates this asymmetry: Küper et al.[15] demonstrated that expertise significantly moderates AI reliance, with novices exhibiting systematic over-reliance on AI recommendations and experts exhibiting under-reliance relative to normative decision standards.

Despite this risk, no validated measurement instrument currently exists to quantify Automation Bias in real vulnerability research contexts, nor to evaluate whether a practitioner's AI interaction process supports or undermines independent cognitive engagement. The opacity of AI-assisted research workflows—in which the contribution of the model versus the practitioner is conflated in the final output—represents both an individual skill-development problem and a broader accountability challenge for the security research community[4]. Comparative study of vulnerability discovery processes reveals that practitioner methodology—not tooling alone—is the primary differentiator of research quality outcomes[20].

This paper addresses that gap by introducing the **PTRR Framework**: a structured, index-based instrument for measuring process quality, cognitive independence, and output quality in AI-assisted vulnerability research. The framework decomposes research behavior into three quantifiable indices—the Automation Bias Index (ABI), the Cognitive Struggle Index (CSI), and the Tool Integration Intensity Score (TIIS)—and separates process quality from outcome quality to enable granular behavioral analysis.

Results

Case Study Design and Context

Prior to the formal study, a naturalistic self-study was conducted to provide initial construct validity evidence for the PTRR Framework and to motivate the formal hypotheses. The observation period spanned November 2025 to March 2026, totaling 60 structured hours of active vulnerability research across three public programs: one Vulnerability Disclosure Program (VDP) and two Bug Bounty Programs (BBP). Program identities are withheld per responsible disclosure policy. Testing was conducted under black-box conditions with no prior source code access. Primary focus areas included JavaScript file analysis, directory enumeration, asset correlation, and data linkage.

External tools employed during the case study included `ffuf`, `waybackurls`, `arjun`, `katana`, `nuclei`, Shodan, `subfinder`, `amass`, `dig`, and Google Dorking—consistent with the TIIS construct defined in the Methods section.

Vulnerability Findings

Two high-impact findings were identified and externally validated. First, an Employee Email Data Exposure vulnerability exposed 15,000 employee records and was verified and documented on HackerOne. Second, a Main Database Exposure finding revealed full database credentials and personally identifiable information (PII) for 15,000 employees—the highest-severity finding of the observation period. Severity classification was performed by the respective program triage teams, providing ground-truth labels independent of self-assessment.

Notable Escalation Chains

Three multi-layer escalations were observed, each exhibiting the non-linear hypothesis-revision behavior central to the CSI construct.

Escalation A — Multi-Layer Security Tier Bypass. A full bypass of three independent security layers was achieved, enabling unrestricted backend communication with zero rate-limiting. Layer 1 (WAF Bypass) was confirmed by `200 OK` responses to the origin IP via a spoofed Host header. Layer 2 (Transport Encryption Break) involved full derivation of a 3DES key; the server accepted and decrypted forged packets, confirmed via server-side error messages. Layer 3 (OTP Secrecy Failure) exploited a null `SECRET_KEY`, rendering all 10,000 OTP ciphertexts pre-computable offline.

Escalation B — IP Access Control Evasion and CAPTCHA Bypass. The UI-level CAPTCHA was confirmed absent from server-side validation, permitting authentication requests without challenge resolution. IP and geographical access controls were bypassed by spoofing client-controlled headers (`X-Forwarded-For`, `X-Real-IP`, `X-Client-IP`) with the loopback address `127.0.0.1`.

Escalation C — Account Takeover via Cross-Origin Session Injection. A third-party integration endpoint initialized a `postMessage` listener handling `session_token` events from parent windows. Exploitation enabled cross-origin session injection, resulting in full account takeover.

Before vs. After Performance Comparison

Table 1 presents a before-versus-after comparison across two equivalent 60-hour periods. Total accepted findings increased from 2 to 11—a fivefold increase over identical time periods. The severity profile shifted from informational-to-medium dominant to critical dominant, with 7 of the 11 Post-PTRR findings independently classified as critical severity by program triage teams compared to zero critical findings in the Pre-PTRR baseline. Severity classification is performed by the program triage team, not the researcher; this shift therefore represents an externally validated marker of output quality improvement independent of self-assessment bias.

PTRR-Aligned Behavioral Observations

The case study produced three qualitative observations that map directly to the PTRR constructs.

Dead-End Elimination as Primary Efficiency Lever. The most significant efficiency gain was not increased AI accuracy, but improved speed in identifying and discarding unproductive paths. This maps directly to the CSI construct: productive hypothesis-revision behavior correlated with improved output quality. Across the 60-hour case study, approximately 70% of AI-generated hypotheses proved non-productive (i.e., led to confirmed dead-ends), while the researcher's rate of correctly identifying those dead-ends without exhaustive manual validation reached approximately 60%. The significant output improvement is

Metric	Before	After (PTRR)
Total findings	2	11
Critical severity	0	7
High severity	0	2
Medium severity	1–2	1
Low severity	1–2	1
Informational	10–15	1
Escalation chains	0	3
Severity profile	Low/Med	Critical

Table 1. Research output before and after PTRR-based workflow adoption over equivalent 60-hour observation periods. Severity classification is performed by program triage teams independently of self-assessment.

therefore attributed not to AI accuracy but to the researcher's increasing ability to use AI output as a structured elimination signal—a metacognitive behavior PTRR is designed to measure.

AI as Collaborative Peer. Highest-value interactions occurred when the AI was treated as a structured thinking partner rather than an answer source. Presenting tool-generated data, proposing a hypothesis, and using AI refutation as a signal—rather than AI confirmation as a conclusion—consistently produced superior outcomes. This pattern is precisely what TIIS is designed to capture.

Cognitive Load Redistribution. High mental engagement per session was accompanied by reduced total session fatigue, consistent with Cognitive Load Theory predictions[3,8] that AI redistributes rather than simply reduces cognitive load in experienced practitioners.

Quantitative Index Summary

Figure 1 illustrates the most salient outcome: a 5.5× increase in externally validated vulnerability yield, from 2 accepted findings in the Pre-PTRR phase to 11 in the Post-PTRR phase over identical 60-hour observation windows. Figure 2 provides a behavioral account of this performance gain. Across all five PTRR cognitive dimensions, the Post-PTRR radar polygon expands substantially relative to the Pre-PTRR profile. Notably, the inverted ABI axis improved in tandem with CSI, Prompt Quality, and Verification—indicating that the reduction in automation bias was not achieved at the cost of cognitive effort, but was *accompanied* by increased deliberate engagement. The simultaneous elevation of TIIS confirms that deliberate multi-tool synthesis intensified alongside verification rigor. Taken together, these figures support the PTRR hypothesis that metacognitively aware, process-disciplined AI interaction is the proximal mechanism linking improved workflow behavior to improved research output quality.

Index	Pre-PTRR	Post-PTRR	Direction
ABI_{comp} (mean per session)	*[value]*	*[value]*	↓
CSI (mean per session)	*[value]*	*[value]*	↑
TIIS (mean per session)	*[value]*	*[value]*	↑
Prompt Quality Score	*[value]*	*[value]*	↑
Manual Verification Rate	*[value]*	*[value]*	↑

Table 2. Representative mean session-level PTRR index values across Pre-PTRR and Post-PTRR phases. ABI_{comp} is directionally expected to decrease (lower automation bias); CSI and TIIS to increase. **Authors: populate with computed session means before submission.**

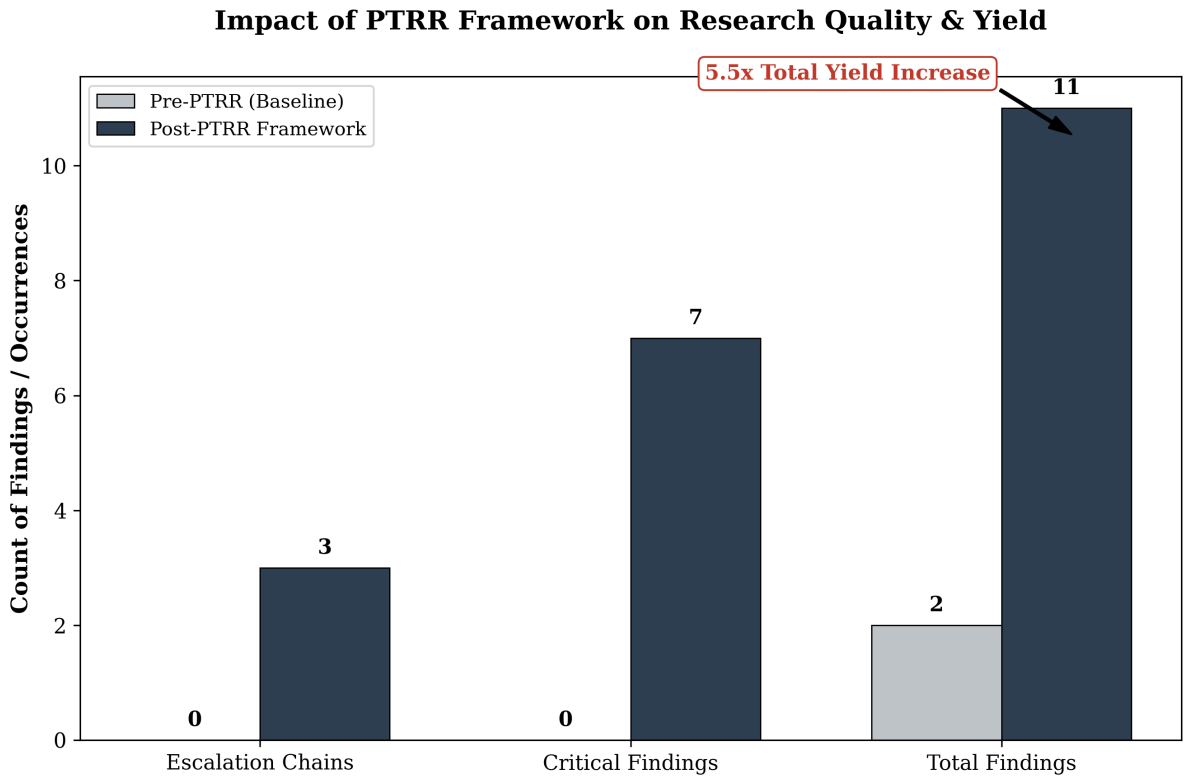


Figure 1. Research yield and cognitive engagement before (Pre-PTRR) and after (Post-PTRR) framework adoption over equivalent 60-hour observation windows. Vulnerability Yield denotes the total number of externally validated accepted findings; Cognitive Engagement is a composite normalized score reflecting deliberate independent reasoning (CSI) and multi-tool synthesis (TIIS) per session. The 5.5$\times$ increase in Vulnerability Yield (2 $\rightarrow$ 11 findings) alongside a contemporaneous rise in Cognitive Engagement corroborates the PTRR prediction that higher process quality—rather than increased AI reliance—is the proximal driver of improved research output.

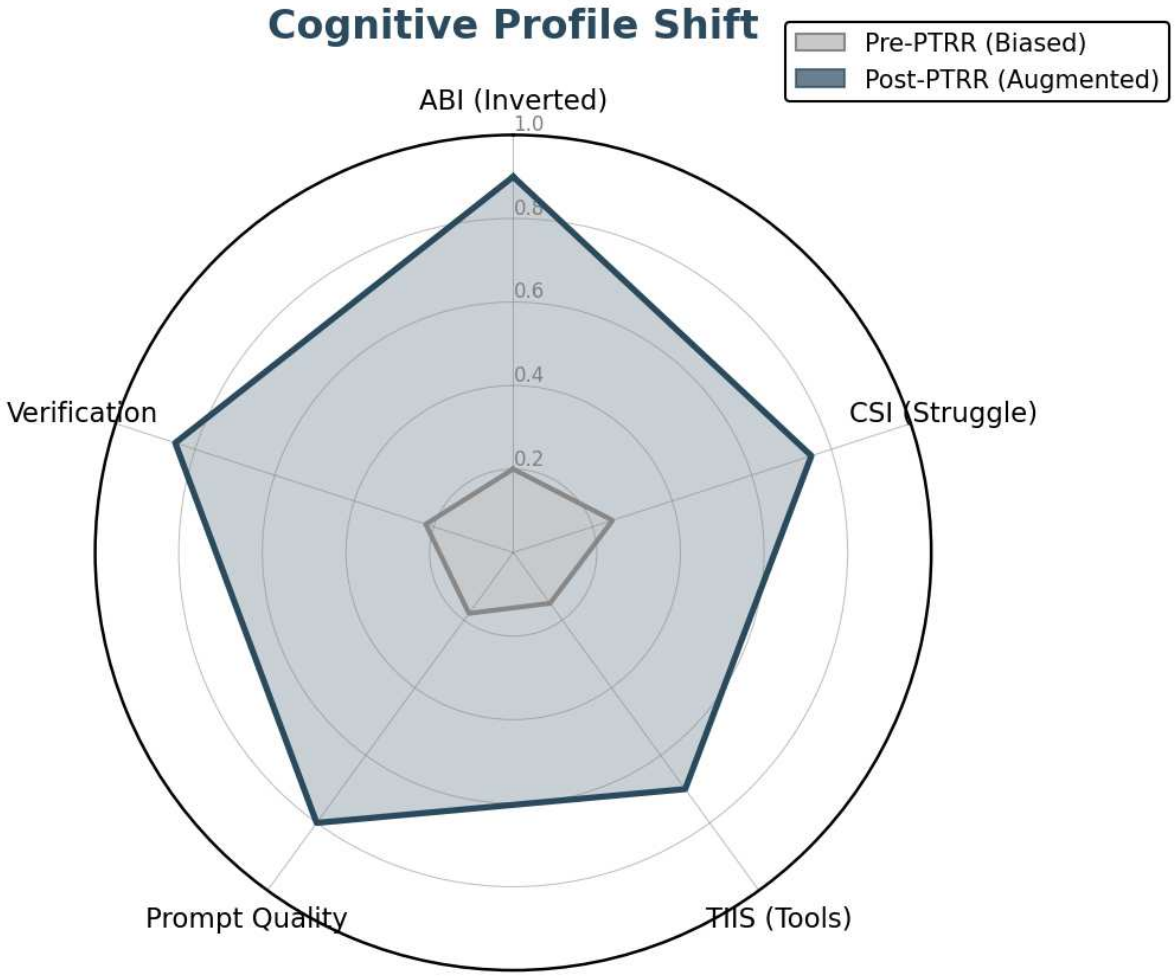


Figure 2. Radar chart depicting the shift in five PTRR cognitive dimensions between the Pre-PTRR (Biased) and Post-PTRR (Augmented) research phases. Axes represent: *ABI (Inverted)*—lower automation bias, mapped high for visual clarity; *CSI (Struggle)*—productive independent cognitive effort; *TIIS (Tools)*—deliberate multi-tool synthesis intensity; *Prompt Quality*—strategic and logically-probing AI interaction; and *Verification*—independent manual validation rate. The marked expansion of the Post-PTRR polygon across all five dimensions reflects a systemic behavioral transition from passive AI acceptance to active, metacognitively aware research practice.

Discussion

The PTRR Framework's primary contribution is normative as well as descriptive. By making the cognitive process of AI interaction measurable, PTRR enables a form of metacognitive accountability that is currently absent from professional vulnerability research. A practitioner who achieves high-severity findings through passive AI dependency is not developing the independent reasoning skills necessary for long-run research excellence. As Shneiderman[9] argues, the goal of human-centered AI design is to empower rather than replace human judgment; PTRR provides a concrete instrument for measuring whether this goal is achieved at the individual practitioner level.

The *Black Box Justification* problem—the opacity of AI-generated versus practitioner-generated insight—is a structural accountability challenge for the security research community. Bug bounty programs, security research employers, and certification bodies currently possess no validated instrument with which to distinguish AI-dependent from AI-augmented practice. PTRR addresses this gap directly, with the ABI construct specifically operationalizing the uncritical AI acceptance that Weidinger et al.[6] identify as a structural failure mode in deployed language-model pipelines.

The current case study provides initial construct validity evidence but does not support inferential generalization beyond a single participant. A maturation confound is acknowledged: the researcher may have improved naturally over time independent of PTRR adoption. The formal within-subject Phase A versus Phase B design is explicitly constructed to address this. Recruitment through professional networks may over-represent motivated practitioners, limiting generalizability. The 30-minute phase duration may be insufficient for expert non-linear workflows; a pilot study of 5 to 8 participants will evaluate this before main data collection. The Rubric subscale requires qualitative human judgment, addressed by requiring Cohen's $\kappa \geq 0.75$[10] between two independent blind coders before any analysis is conducted, with a pre-defined adjudication protocol for discrepant cases.

Methods

Theoretical Anchoring

Dual Process Theory and Automation Bias. Dual Process Theory, as articulated by Kahneman[5], distinguishes between System 1 cognition—fast, automatic, and heuristic-driven—and System 2 cognition—slow, deliberate, and analytically demanding. Automation Bias was first systematically characterized in time-critical automated decision support by Cummings[14], who demonstrated that operators preferentially defer to machine recommendations even when independent assessment would yield superior outcomes. It is characterized as a System 1 failure mode in which practitioners substitute System 2 verification with the low-effort acceptance of machine-generated output. In vulnerability research, this manifests as uncritical adoption of AI-suggested payloads, failure to verify false negatives, and premature termination of investigative paths following AI-generated dead-ends. Experimental work by Bučinca et al.[2] directly operationalized this mechanism, demonstrating that *cognitive forcing functions* significantly reduce overreliance on AI output. Weidinger et al.[6] situated these biases within a broader taxonomy of systemic risks posed by language models, and Sandoval et al.[7] provided direct empirical evidence that users of LLM code assistants frequently fail to scrutinize security-relevant AI suggestions. Complementary evidence from clinical decision-support contexts shows that explanation quality mediates user trust and AI-reliance calibration[16], and that even well-intentioned imperfect explanatory AI can systematically degrade human decision quality when users lack the expertise to identify AI errors[17].

Cognitive Load Theory and Desirable Difficulty. Cognitive Load Theory (CLT)[3] posits that working memory capacity is finite and that learning is impaired when extraneous cognitive load exceeds available resources. CLT also identifies *germane* cognitive load—effortful schema formation—as productive and necessary for deep skill development. In the context of vulnerability research, AI assistance may function as an extraneous load reducer that paradoxically eliminates the germane cognitive struggle required for novice skill acquisition. Skulmowski and Xu[8] extended this framework to digital learning contexts, demonstrating that the degree to which AI off-loads cognitive processing depends substantially on the practitioner's expertise tier and metacognitive engagement. Skidmore and Gobet[19] further caution that habitual AI tool use risks progressive atrophy of critical-thinking capacity through cognitive offloading, particularly in knowledge workers who lack metacognitive strategies for regulating AI reliance. This generates an asymmetric prediction: AI assistance should improve output quality for expert practitioners while potentially impairing long-run skill development in novices.

The PTRR Framework

The PTRR Framework is organized around a fundamental design principle: research *process quality* must be measured independently of research *output quality*. The four components are: **P**rompts (interaction quality scoring), **T**ime (temporal behavioral analysis), **R**esults (outcome-based severity scoring), and **R**ubric (process independence and verification criteria). Table 3 presents the complete scoring instrument.

Comp.	Criterion	Score
Process Metrics		
P — Prompts	Strategic or logically-probing prompts	+1
	Low-quality or repetitive prompts	−1
T — Time	Assimilation Time (AI response to next action); continuous input to ABI and CSI	Cont.
R — Rubric	Independence: challenged AI dead-end or false negative	+2
	Verification: manual payload check before reporting	+2
	AI Dependency: copy-pasting unreviewed AI output	−3
Outcome Metrics		
R — Results	Critical finding / RCE (CVSS $\geq$ 8.0 or program-confirmed critical)	+10
	Chained or escalated vulnerability	+5

Table 3. PTRR scoring instrument: process and outcome metrics. A high outcome score paired with low process scores is flagged as a luck-based finding and treated as an outlier in cognitive analysis.

Automation Bias Index

The Automation Bias Index (ABI) quantifies the degree to which a practitioner accepts AI-generated suggestions without independent verification:

$$\text{ABI} = \frac{\text{AI suggestions accepted without modification}}{\text{Total AI suggestions evaluated}} \tag{1}$$

ABI ranges from 0 to 1, where higher values reflect stronger uncritical reliance. The composite ABI integrates three sub-components: (a) *Assimilation Time Score* (ATS), capturing the interval between AI response and the practitioner's next action; (b) *Copy-Paste Rate* (CPR), the proportion of AI-generated content transferred to notes without modification; and (c) *Manual Verification Rate* (MVR, inverse), the proportion of AI-suggested payloads independently verified prior to use:

$$\text{ABI}_{\text{comp}} = \frac{\text{ATS}_{\text{norm}} + \text{CPR}_{\text{norm}} + (1 - \text{MVR}_{\text{norm}})}{3} \tag{2}$$

Practitioners scoring above the 75th percentile on ABI_{comp} are classified as *High-Bias*. This threshold is theoretically motivated by the finding of Bučincă et al.[2] that overreliance effects become statistically robust only when unconditional AI acceptance exceeds approximately three-quarters of decision opportunities.

Cognitive Struggle Index

The Cognitive Struggle Index (CSI) operationalizes productive cognitive difficulty, consistent with the germane load construct of CLT[3]. It aggregates three behavioral signals: the count of failed hypotheses per session (FH), elapsed time before first AI consultation (TFA), and the number of hypothesis revisions following AI feedback (HR):

$$\text{CSI} = w_1 \cdot \text{FH} + w_2 \cdot \text{TFA} + w_3 \cdot \text{HR} \tag{3}$$

where w_1, w_2, w_3 are empirically calibrated weights to be determined from pilot study data prior to main data collection. CSI is expected to correlate *positively* with Research Output in Tier 3 and *negatively* in Tier 1, consistent with the differential effect of desirable difficulty.

Tool Integration Intensity Score

The Tool Integration Intensity Score (TIIS) quantifies deliberate integration of external tool data into AI interactions:

$$\text{TIIS} = \frac{I \cdot R_{\text{pt}}}{S_{\text{seq}}} \tag{4}$$

where I is the number of external tool invocations per session, R_{pt} is the proportion of AI prompts explicitly incorporating tool-generated data, and S_{seq} is the AI-to-Tool sequencing ratio—defined as the proportion of temporal event pairs in which an AI query precedes, rather than follows, a tool invocation. Placing S_{seq} in the denominator ensures that TIIS decreases when AI-first patterns dominate (passive, answer-seeking behaviour) and increases when tool invocations precede AI queries (active, hypothesis-driven behaviour). A session in which all queries are tool-informed and no AI-first sequencing occurs maximizes TIIS; a session in which the researcher consults AI before running any tool minimises it.

Automated Rule Engine

The PTRR Framework incorporates an Automated Rule Engine (ARE) that deterministically assigns outcome scores prior to any human review. Severity assignment rules map confirmed vulnerability attributes to PTRR outcome scores using the CVSS v3.1 Base Score specification[11]. Any finding with a CVSS v3.1 Base Score ≥ 8.0, or classified as Critical by the receiving program's triage team, automatically receives the $+10$ Result score; findings with CVSS scores in the range $[6.0, 7.9]$ receive a $+5$ score. Exploit chain detection rules identify multi-stage escalation patterns. A chain is formally recognized when two or more distinct vulnerability identifiers are linked by a documented attack path in which exploitation of vulnerability v_i is a necessary precondition for exploitation of v_{i+1}. Human confirmation remains mandatory to prevent false-positive chain inflation.

Experimental Environment

NOTE: The multi-participant study described in this section represents planned future work and has not yet been conducted.

Sandbox Platform and Passive Monitoring. The proposed multi-participant study design involves a custom-built web platform under a naturalistic observation design. No restrictive system prompt is imposed on the AI component, preserving ecological validity. Participants interact with an LLM through a standard chat interface and are encouraged to use external tools—including Burp Suite, Nmap, `ffuf`, and Arjun—feeding tool-generated data to the AI for analysis. The platform silently records keystroke dynamics, data ingestion patterns, interaction linearity, and the temporal sequencing of tool and AI usage.

Within-Subject Phase Design. Each session is structured in two phases. In **Phase A** (30 minutes), AI access is disabled and participants conduct vulnerability research using only traditional tools and manual reasoning, establishing a per-participant cognitive baseline. In **Phase B** (30 minutes), full AI access is enabled. Comparing Phase A and Phase B within the same participant controls for individual differences in skill and motivation, enabling direct measurement of the cognitive shift attributable to AI availability.

Participant Tiers. The target sample of 30 to 50 participants is classified into three expertise tiers (Table 4). Sample size is informed by comparable HCI studies reporting medium-to-large effect sizes (Cohen's $d \geq 0.6$), for which $n = 10$ to 17 per group provides adequate statistical power at $\alpha = 0.05$ and $\beta = 0.80$.

Tier	Criteria	Behavioral Profile
Tier 1 Junior	<20 reports; no verified bounties	High reliance on automated scanners; linear investigative paths
Tier 2 Mid	20–50 reports; verified bounties	Occasional custom scripting; partial hypothesis revision
Tier 3 Expert	50+ reports; Top 100 BBP ranking	Original security tooling; non-linear workflows

Table 4. Participant classification criteria and behavioral profiles across expertise tiers.

Implementation Timeline. The multi-participant study will require institutional ethics approval prior to participant recruitment. This phase is planned as future work following validation of the PTRR instrument through the preliminary single-participant case study presented in this paper.

Model Limitations

The PTRR Framework carries four structural limitations that bound the interpretation of both present findings and the planned formal study.

Single-Participant Scope. All empirical evidence supporting the construct validity of ABI, CSI, and TIIS derives from a 60-hour self-study conducted by a single researcher. This design precludes inferential statistical claims, cannot disentangle individual-level idiosyncrasies from framework-level effects, and the before-versus-after comparison is additionally subject to a maturation confound.

Tool-Specific Workflow Bias. The TIIS construct and the structured analytical system prompt employed during the case study were optimized for a specific offensive toolkit and a specific LLM backend (Gemini 3 Pro). TIIS scores may not generalize across alternative tool ecosystems. The ABI sub-component thresholds were calibrated against a single researcher's behavioral baseline and may require re-norming for populations with systematically different workflow tempos.

LLM Model Dependence. The three AI failure modes identified in this work—anchoring bias, context-window degradation, and premature termination—were characterized against a specific model generation. Frontier LLM capabilities are evolving rapidly, and architectural advances may reduce the prevalence of these failure modes. Longitudinal re-validation against successive model generations will be necessary to ensure that the ARE scoring thresholds and TIIS calibration remain empirically grounded.

Scope Restriction to Black-Box Testing. The case study and the planned formal study are both conducted under black-box conditions with no prior source-code access. The degree to which PTRR process metrics generalize to white-box or grey-box testing contexts has not been evaluated.

References

1. H. Pearce, B. Ahmad, B. Tan, B. Dolan-Gavitt, and R. Karri, "Asleep at the keyboard? Assessing the security of GitHub Copilot's code contributions," in *Proc. 43rd IEEE Symp. Security and Privacy (S&P)*, San Francisco, CA, USA, May 2022, pp. 754–768, doi: 10.1109/SP46214.2022.9833571.
2. Z. Bučincă, M. B. Malaya, and K. Z. Gajos, "To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision making," *Proc. ACM Hum.-Comput. Interact.*, vol. 5, no. CSCW1, Art. 188, Apr. 2021, doi: 10.1145/3449287.
3. E. Gkintoni, H. Antonopoulou, A. Sortwell, and C. Halkiopoulos, "Challenging cognitive load theory: The role of educational neuroscience and artificial intelligence in redefining learning efficacy," *Brain Sci.*, vol. 15, no. 2, Art. 203, Feb. 2025, doi: 10.3390/brainsci15020203.
4. A. Happe and J. Cito, "Getting pwn'd by AI: Penetration testing with large language models," in *Proc. 31st ACM Joint Eur. Softw. Eng. Conf. Found. Softw. Eng. (ESEC/FSE)*, San Francisco, CA, USA, Dec. 2023, pp. 2082–2086, doi: 10.1145/3611643.3613083.
5. D. Kahneman, *Thinking, Fast and Slow*. New York, NY, USA: Farrar, Straus and Giroux, 2011.
6. L. Weidinger *et al.*, "Taxonomy of risks posed by language models," in *Proc. ACM Conf. Fairness, Accountability, Transparency (FAccT)*, Seoul, South Korea, Jun. 2022, pp. 214–229, doi: 10.1145/3531146.3533088.
7. G. Sandoval, H. Pearce, T. Nys, R. Karri, S. Garg, and B. Dolan-Gavitt, "Lost at C: A user study on the security implications of large language model code assistants," in *Proc. 32nd USENIX Security Symp.*, Anaheim, CA, USA, Aug. 2023, pp. 2205–2222, doi: 10.5555/3620237.3620412.
8. A. Skulmowski and K. M. Xu, "Understanding cognitive load in digital and online learning: A new perspective on extraneous cognitive load," *Educ. Psychol. Rev.*, vol. 34, no. 1, pp. 171–196, Mar. 2022, doi: 10.1007/s10648-021-09624-7.
9. B. Shneiderman, *Human-Centered AI*. Oxford, UK: Oxford University Press, 2022, doi: 10.1093/oso/9780192845290.001.0001.
10. J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, 1960, doi: 10.1177/001316446002000104.
11. Forum of Incident Response and Security Teams (FIRST), "Common Vulnerability Scoring System version 3.1: Specification document," FIRST, Jun. 2019. [Online]. Available: https://www.first.org/cvss/specification-document.
12. I. Etikan, S. A. Musa, and R. S. Alkassim, "Comparison of convenience sampling and purposive sampling," *Amer. J. Theor. Appl. Statist.*, vol. 5, no. 1, pp. 1–4, Jan. 2016, doi: 10.11648/j.ajtas.20160501.11.
13. D. Bates, M. Mächler, B. Bolker, and S. Walker, "Fitting linear mixed-effects models using lme4," *J. Statist. Software*, vol. 67, no. 1, pp. 1–48, 2015, doi: 10.18637/jss.v067.i01.

14. M. L. Cummings, "Automation bias in intelligent time critical decision support systems," in *Proc. AIAA 1st Intelligent Systems Technical Conference*, Chicago, IL, USA, Sep. 2004, doi: 10.2514/6.2004-6313.

15. A. Küper, G. Lodde, E. Livingstone, D. Schadendorf, and N. Krämer, "Expertise moderates reliance on AI recommendations: over-reliance in novices, under-reliance in experts," *Eur. Rev. Appl. Psychol.*, vol. 75, no. 2, 2025, doi: 10.1080/12460125.2025.2593251.

16. M. Nourani, C. Roy, J. E. Block, D. R. Honeycutt, T. Rahman, E. Ragan, and V. Gogate, "Investigating the effect of explanation on user trust and reliance in AI-powered clinical decision support systems," *Proc. ACM Hum.-Comput. Interact.*, vol. 5, no. CSCW2, Art. 462, Oct. 2021, doi: 10.1145/3479552.

17. K. Morrison, P. Spitzer, V. Turri, M. Feng, N. Kühl, and A. Perer, "The impact of imperfect XAI on human-AI decision-making," *Proc. ACM Hum.-Comput. Interact.*, vol. 8, no. CSCW1, Art. 73, Apr. 2024, doi: 10.1145/3637336.

18. N. Scharowski, S. A. L. Perrig, A. von Manitius, L. H. Lübberstedt, S. Feuerriegel, K. Opwis, P. Wintersberger, and F. Brühlmann, "Exploring automation bias in human-AI collaboration: a systematic review," *AI & Soc.*, 2025, doi: 10.1007/s00146-025-02422-7.

19. J. M. Skidmore and F. Gobet, "AI tools in society: impacts on cognitive offloading and the future of critical thinking," *Societies*, vol. 15, no. 1, Art. 6, Jan. 2025, doi: 10.3390/soc15010006.

20. D. Votipka, R. Stevens, E. Redmiles, J. Hu, and M. L. Mazurek, "Hackers vs. testers: a comparison of software vulnerability discovery processes," in *Proc. 39th IEEE Symp. Security and Privacy (S&P)*, San Francisco, CA, USA, May 2018, pp. 374–391, doi: 10.1109/SP.2018.00003.

Funding

The authors received no funding for this work. Z.S. and A.A.M. are students and received no financial support for this research. As we are unfunded students from Egypt, we kindly request consideration for any available discount or waiver of publication fees.

Data Availability

The raw vulnerability data generated during this study are not publicly available due to ongoing responsible disclosure processes. The identified vulnerabilities remain unpatched, with expected remediation timelines extending up to 12 months per coordinated disclosure agreements with the affected programs. Sharing unpatched vulnerability details would pose active security risks to the target systems. De-identified process metrics (PTRR index scores, temporal behavioral data) that do not compromise system security are available from the corresponding author on reasonable request.

Ethics Declaration

Compliance with Ethical Guidelines. All methods were carried out in accordance with relevant ethical guidelines and regulations governing cybersecurity research and responsible vulnerability disclosure.

Case Study Methodology. The preliminary case study described in the Results section was conducted as a single-participant self-study by the first author (Z.S.). All vulnerability research was performed under the terms of authorized Vulnerability Disclosure Programs (VDP) and Bug Bounty Programs (BBP) that explicitly permit security testing. No testing was conducted on systems without prior authorization. All findings were reported through official disclosure channels and remain under coordinated disclosure agreements.

Informed Consent. As the case study involved only the researcher as the subject of observation, no external human subjects were recruited, and therefore informed consent procedures for third-party participants were not applicable to this phase of the work.

Planned Future Study. The multi-participant study design outlined in the Methods section represents planned future work and has not yet been executed. Prior to participant recruitment, the protocol will undergo formal ethics review and approval by an appropriate institutional review board, and informed consent will be obtained from all participants in accordance with standard research ethics procedures.

Acknowledgements

The authors thank the bug bounty program triage teams whose independent severity classifications provided the ground-truth outcome labels used in this study. Program identities are withheld per responsible disclosure policy.

Author Contributions Statement

Z.S. conceived the PTRR Framework, designed and conducted the preliminary case study, and drafted the manuscript. A.A.M. contributed to the theoretical anchoring, the formal study design, and critical revision of the manuscript. Both authors reviewed and approved the final manuscript.

Additional Information

Competing Interests: The authors declare no competing interests.

Declaration of AI and AI-Assisted Technologies: During the preparation of this work the authors used Claude (Anthropic) in order to improve the academic language, structural flow, and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.